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Project Work - Artificial Vision

Group 23

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1. Introduction 3
1.1. Description of the problem 3
1.2. Possible solutions 4
2. Dataset description 5
2.1. Dataset composition 5
2.2. Annotations and Ground Truth 5
2.3. Dataset Pre-Processing and Formatting 6
2.4. Re-Identification Dataset Generation 6
3. Player Detection 7
3.1 Model Selection 7
3.2 Training Strategy and Optimization 8
3.3 Performance Analysis and Model Validation 8
4. Player Tracking 11
4.1. Tracking Model Selection 11
4.2 Re-Identification Strategy & Model Fine-Tuning 12
4.3 Optimization Strategy 12
4.4 Offline Post-Processing Pipeline 13
4.5 Qualitative Validation 14
5. Behavior Analysis 14
5.1 Methodological Approach 15
5.2 Performance Analysis 15
6. Discussion and Conclusions 15

1. Introduction

The field of sports analytics has been significantly transformed by recent advancements in Artificial Intelligence and Computer Vision. Automated systems capable of interpreting game dynamics from video footage are increasingly in demand for tactical analysis, broadcasting, and scouting. This report presents the methodology and results developed for the Soccer Video Understanding contest organized by the MIVIA Lab and the Department of Information Engineering (DIEM) at the University of Salerno. The primary goal of this project is to design a computer vision pipeline capable of automatically extracting high-level semantic information from raw soccer footage recorded via moving cameras.

1.1. Description of the problem

Contest Overview

The contest challenges participants to address two distinct but complementary tasks within the domain of soccer video analysis: Player Tracking and Player Behavior Analysis. The system must operate on a test set consisting of 5 video clips, each lasting 30 seconds and acquired with a resolution of 1920 × 1080 pixels.

Domain Challenges

In contrast to static surveillance scenarios, soccer footage presents a unique set of challenges that render traditional approaches inadequate:

- **Camera Dynamics:** Broadcast or sideline footage entails rapid camera movements (panning, tilting, zooming) that render traditional background subtraction techniques ineffective and introduce significant motion blur.
- **Scale Variation:** Depending on the zoom level, a player's spatial resolution can range from a few dozen to several hundred pixels. Detecting players on the far side of the pitch demands extremely high sensitivity to fine-grained details.
- **Occlusions and Deformations:** Players frequently move in clusters, causing mutual occlusions, and assume non-rigid poses (e.g., running, sliding, jumping). These factors complicate identity preservation, often leading to ID Switches.
- **Semantic Ambiguity:** The system must effectively distinguish active players from visually similar distractors, such as substitutes on the bench, coaching staff, or spectators in the stands.

Task Specifications

- **Player Detection and Tracking:** The first objective is to detect and re-identify players in real-time from the video stream. For each video, the system is required to generate a tracking file (*tracking_K_XX.txt*) detailing the frame identifier, object identifier, and the bounding box coordinates (*top-left x, top-left y, width, and height*) for every player instance. The performance for this task is evaluated using the HOTA0.5 metric, which balances detection accuracy ($DetA_{0.5}$) and association accuracy ($AssA_{0.5}$).
- **Player Behavior Analysis:** The second objective focuses on estimating the density of players within specific areas of the pitch. For each video, two rectangular Regions of Interest (*ROIs*) are defined via configuration files containing relative coordinates. The system must estimate the number of players inside these ROIs for every frame. A player is considered to be inside a region if the center of the base of their bounding box falls within the ROI boundaries. The error for this task is measured using the normalized Mean Absolute Error (*nMAE*).

The final ranking of the contest is determined by the Player Tracking and Behavior Score (*PTBS*), calculated as the sum of the tracking performance and the behavior analysis score:

$$PTBS = HOTA_{0.5} + nMAE$$

1.2. Possible solutions

Addressing these challenges requires a sophisticated pipeline, typically structured around the Tracking-by-Detection paradigm. Given the dynamic nature of the footage and the moving camera, traditional background subtraction methods are ineffective. Instead, state-of-the-art Deep Learning approaches are necessary.

A standard solution involves a high-performance Object Detector (such as the YOLO family) to localize players in each frame, followed by a Multi-Object Tracking (MOT) algorithm. The tracker must leverage motion models (e.g., Kalman Filters) and appearance descriptors (Re-Identification embeddings) to maintain consistent identities despite frequent occlusions and erratic player movements. Furthermore, the behavior analysis task can be formulated as a geometric post-processing step: by mapping the tracked bounding boxes to the image coordinates, the system can determine inclusion within the specified ROIs provided in the dataset configuration. The specific architectural choices and experimental strategies adopted to maximize the PTBS are detailed in the subsequent sections of this report.

2. Dataset description

To develop and evaluate the proposed tracking pipeline, we utilized the SoccerNet Tracking benchmark, a large-scale dataset specifically designed for Multi-Object Tracking (MOT) in soccer videos. Unlike generic tracking datasets, SoccerNet focuses on the unique challenges of sports analytics, such as rapid camera movements, similar player appearances, and frequent occlusions.

2.1. Dataset composition

The SoccerNet Tracking dataset is derived from 12 complete soccer games captured from the main broadcast camera view. The footage is recorded in Full HD resolution (1920 × 1080 pixels), providing high-fidelity visual information necessary for detecting small o distant players.

For the purpose of this project, we focused exclusively on the officially defined Train and Test splits provided by the benchmark authors:

- **Training Set:** Consists of 57 video clips.
- **Test Set:** Comprises 49 video clips.

Each individual video clip has a fixed duration of 30 seconds. To facilitate frame-by-frame analysis, the videos are decomposed into discrete image sequences. Specifically, each clip consists of 750 frames, corresponding to a frame rate of 25 fps. This dense temporal sampling is critical for the tracking task, as it allows the model to capture fine-grained motion patterns and maintain identity consistency even during rapid player movements.

2.2. Annotations and Ground Truth

The dataset provides dense annotations for every frame, following standard MOT formats. The ground truth includes:

- **Bounding Boxes:** Defined by the top-left coordinates (*x, y*), width (*w*), and height (*h*) for each object.
- **Track IDs:** Unique identifiers assigned to each distinct object (players, referees, and balls) to support trajectory analysis and re-identification tasks.
- **Class Labels:** Categorization of objects into three primary classes: *Player*, *Goalkeeper*, *Referee*, and *Ball* (though our primary focus remained on player tracking).

The data is structured in comma-separated value files, where each line represents a single object instance in a specific frame (e.g., *Frame ID, Track ID, x, y, w, h*). This standardized structure facilitated the direct integration of the data with state-of-the-art tracking algorithms.

2.3. Dataset Pre-Processing and Formatting

To prepare the SoccerNet dataset for training the YOLOv11l detector, a systematic pre-processing pipeline was implemented to handle data extraction, splitting, and format conversion.

The raw dataset, initially stored as zip archives (train.zip, test.zip) on Google Drive, was extracted into a temporary local environment to optimize I/O operations. A custom Python script was developed to manage the file structure, ensuring a flat hierarchy free of redundant nesting. Specifically, the test set was partitioned to create a dedicated validation split: the first 30 sequences (SNMOT-060 to SNMOT-089) were isolated as the validation set, while the remaining sequences were reserved for testing.

Following extraction, the data was converted into the YOLO format. This process involved parsing the gt.txt (Ground Truth) and gameinfo.ini files for each video sequence to map track IDs to specific classes. A strict filtering logic was applied to retain only relevant entities (players, goalkeepers, and referees) and merge them into a single "person" class (Class ID 0), while excluding irrelevant objects such as the ball. Bounding box coordinates were converted from the MOT format $[x_{top_left}, y_{top_left}, width, height]$ to the normalized YOLO format $[x_{center}, y_{center}, width, height]$. To reduce data redundancy and training time, a sampling stride of 5 frames was applied to both training and validation sets. Finally, a data.yaml configuration file was generated to define the dataset structure for the Ultralytics training framework.

2.4. Re-Identification Dataset Generation

To fine-tune the OSNet architecture for the specific challenges of soccer player re-identification, a custom dataset was generated from the SoccerNet training sequences. A specialized data processing pipeline was developed to extract player crops while filtering out noise and non-human objects.

The raw video sequences were first partitioned into a training set (85% of sequences) and a validation set (15%). For every sequence, the ground truth annotations were parsed to extract bounding boxes for each tracked identity. To ensure the quality of the training data, several filtering criteria were applied:

1. **Class Filtering:** Only bounding boxes labeled as "pedestrian" (Class ID 1) were retained.
2. **Artifact Removal:** Track IDs associated with the "ball" were identified by parsing the gameinfo.ini configuration files and explicitly excluded from the dataset.
3. **Geometric Filtering:** Bounding boxes with an area smaller than 1000 pixels or an anomalous aspect ratio were discarded to remove partial occlusions or tracking errors.

4. **Temporal Sampling:** To minimize data redundancy and prevent overfitting on identical frames, images were extracted with a sampling interval of 10 frames.

The resulting dataset was organized following the standard Market-1501 directory structure to ensure compatibility with the Torchreid library. The training set consists of folders containing all valid crops for each identity found in the training sequences. For the validation sequences, a "Query/Gallery" split was simulated: for each identity, one random image was assigned to the **Query** set, while the remaining images were allocated to the **Gallery** set. This structure allows for the evaluation of the model's ability to retrieve the correct identity from a large pool of candidates.

3. Player Detection

Player detection is the critical foundation of our *Tracking-by-Detection* architecture, implemented using the Ultralytics framework. Since the tracker relies exclusively on frame-by-frame bounding boxes, high detection quality is paramount to prevent failures from propagating downstream. The *SoccerNet Challenge* presents specific hurdles compared to standard datasets: players often appear as small objects occupying less than 0.1% of the frame due to long-shot angles, requiring high-resolution processing. Additionally, rapid actions introduce significant motion blur that complicates boundary regression, while frequent static and dynamic occlusions demand a model with high sensitivity to maintain trajectories.

3.1 Model Selection

The architecture selected for the detection module is YOLOv11, specifically the Large (yolo11l) variant. This choice was driven by the model's architectural evolution, which offers a superior trade-off between inference latency and detection accuracy compared to its predecessors (YOLOv8/v9/v10).

Key architectural properties relevant to this project include:

- **C3k2 Backbone & Neck:** YOLO11 introduces the C3k2 block, a refined iteration of the CSP bottleneck. This enhances feature aggregation, crucial for distinguishing player limbs from the textured grass background.
- **C2PSA (Cross-Stage Partial Spatial Attention):** This spatial attention mechanism enables the model to focus adaptively on relevant features while suppressing background noise. This is particularly effective in reducing false positives in crowded stands ("background clutter").
- **Small Object Optimization:** The re-parameterized architecture maintains high representational power even for small objects. Given that players in $1920 \times$

1080 footage often occupy fewer than 30×30 pixels, this ability to preserve fine-grained features was a deciding factor.

3.2 Training Strategy and Optimization

The training strategy was specifically designed to adapt the model to the unique constraints of the soccer domain and the available hardware resources.

Key experiments and final configurations include:

- **High-Resolution Input:** Initial tests at decreased resolutions (1280px) yielded insufficient recall for players positioned in the background. To address this, we enforced a native input resolution of 1920px. Although computationally demanding, this high-resolution input was strictly necessary to preserve the spatial details of small objects, ensuring they were not lost during downsampling.
- **Batch Size:** Utilizing a 1920px resolution on an L4 GPU (24GB VRAM), we configured a batch size of 8 to balance memory usage with training stability. To further smooth gradients, we combined this with an extended warmup phase (3 epochs).
- **Hyperparameters:** The model was trained for a max of 70 epochs using the AdamW optimizer with a Cosine Learning Rate Scheduler to ensure smooth convergence. An Early Stopping mechanism (patience=10) was employed to prevent overfitting. Furthermore, Mosaic and MixUp augmentations were enabled to improve robustness against partial occlusions, a critical factor during crowded gameplay scenarios.

3.3 Performance Analysis and Model Validation

We conducted a rigorous quantitative analysis on the validation split to assess the model's convergence and generalization capabilities.

- **Training Stability and Convergence:** The training dashboard (Fig 1) confirms that the model converged effectively without signs of overfitting. Both the box regression loss (localization) and classification loss (identification) show a steady decrease, aligning well with the validation curves. Notably, the training process automatically terminated at epoch 48 due to the Early Stopping mechanism, indicating that the model had reached optimal generalization on the validation set, rendering further epochs redundant.

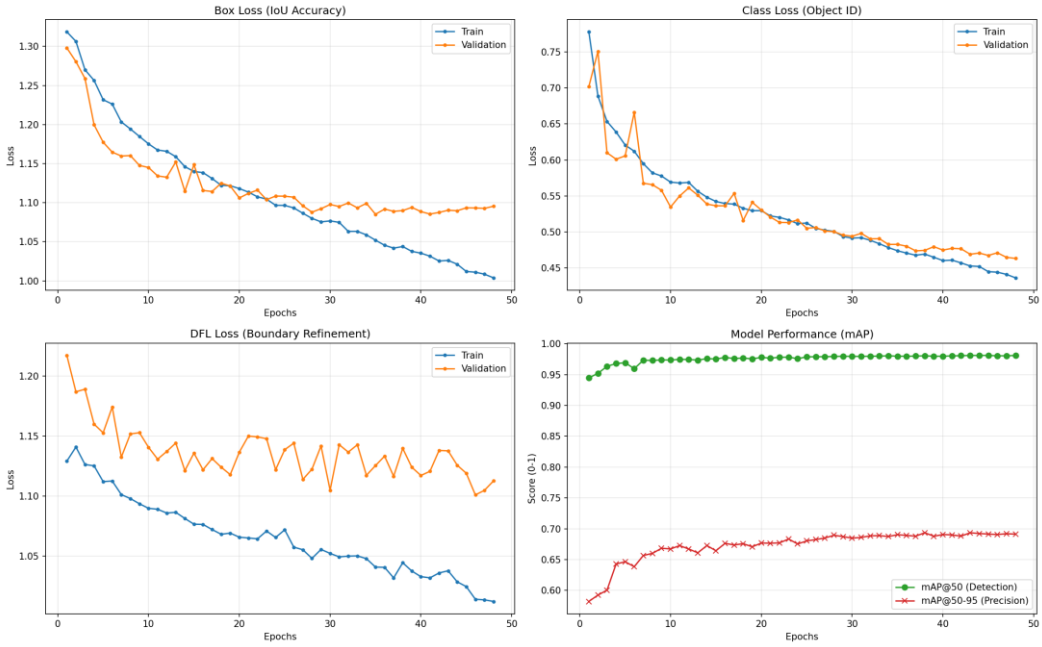


Fig. 1: Training Metrics Dashboard

- **Precision-Recall Robustness:** The Precision-Recall curve (Fig 2) demonstrates high precision even at higher recall levels. The curve approaches the top-right corner (high AUC), confirming that the detections fed into the tracker are reliable, minimizing the risk of "ghost tracks" generated by false positives.

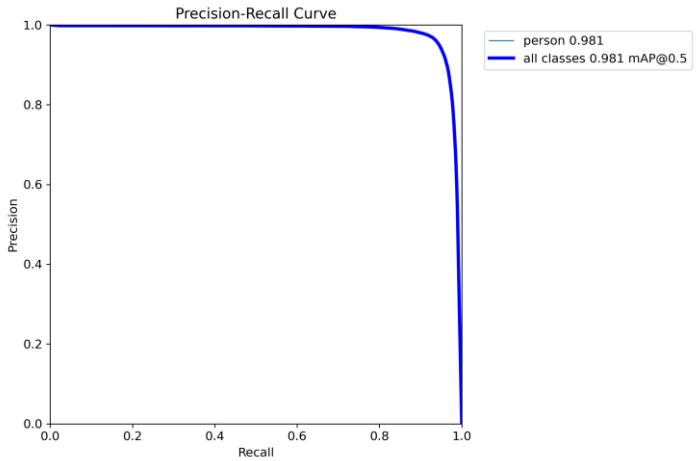


Fig. 2: Precision-Recall Curve

- **Background Discrimination:** The normalized confusion matrix shows strong performance on the ‘person’ class, with limited confusion in player detection, while background misclassifications are handled in the tracking stage.

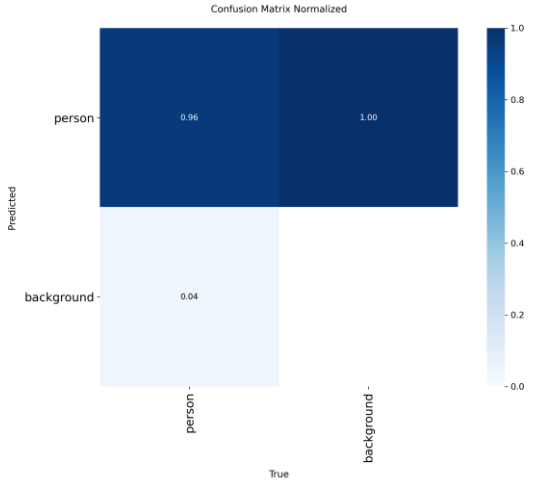


Fig. 3: Normalized Confusion Matrix

- **Recall-Confidence Trade-off:** The Recall-Confidence curve (Fig 4) justifies our inference strategy. To ensure tracking continuity, we selected a low operating point of 0.20, prioritizing Recall to minimize missed detections of small or occluded players.

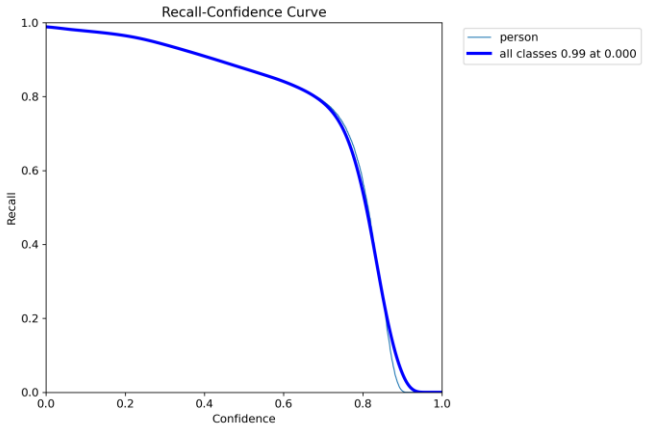


Fig. 4: Recall-Confidence Curve.

- **F1-Score Analysis:** The F1-Confidence curve (Fig 5) indicates the optimal balance between precision and recall. While the peak F1 suggests a higher threshold, our tracking-oriented strategy prioritizes recall (left side of the peak) to avoid losing tracks.

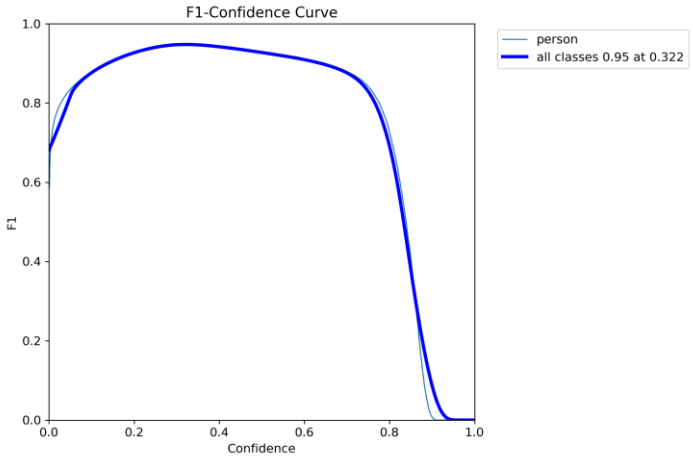


Fig. 5: F1-Score vs Confidence.

4. Player Tracking

Player tracking involves associating detections across consecutive frames to reconstruct the individual trajectories of players, referees, and staff. The primary challenge in soccer is the non-linear motion of these targets combined with rapid camera movement (pan/tilt/zoom).

4.1. Tracking Model Selection

The definition of the tracking architecture followed an iterative evaluation process to address specific failure modes observed in the dataset.

We initially evaluated ByteTrack as a baseline due to its computational efficiency. While effective for static surveillance, ByteTrack proved unstable in dynamic broadcast scenarios: rapid camera pans caused the Kalman Filter's linear prediction to diverge from the actual player position, leading to frequent identity switches.

To resolve this, we transitioned to BoTSORT (Boosted Tracker + SORT). The selection was driven by specific algorithmic advantages crucial for sports analytics:

- **Global Motion Compensation (GMC):** BoTSORT integrates a module based on Sparse Optical Flow to estimate camera movement. By aligning detections into a stabilized coordinate system before association, it neutralizes camera pans, allowing the tracker to correctly maintain IDs even when the player is stationary relative to the field but moving in the frame.
- **Dual Association Metric:** Unlike the motion-only baseline, BoTSORT enhances robustness by fusing motion consistency with deep visual embeddings (ReID).

This hybrid approach provides a secondary mechanism to recover identities after long-term occlusions where spatial prediction fails.

Architectural Distinction (Online vs. Offline): While BoTSORT is intrinsically an Online tracker, our final pipeline operates in an Offline (Batch) mode. The inclusion of a comprehensive post-processing stage (Stitching, Interpolation, Smoothing) requires access to future frames to close trajectory gaps and refine tracks. We deliberately chose this hybrid approach to maximize the PTBS score, prioritizing accuracy over latency.

4.2 Re-Identification Strategy & Model Fine-Tuning

Effective re-identification is critical for recovering identities after prolonged occlusions. To address the specific challenge of distinguishing teammates wearing identical uniforms, we implemented a domain-specific fine-tuning strategy.

- **Model Training (Torchreid):** We fine-tuned an OSNet (Omni-Scale Network) architecture using the `Torchreid` library. The model was trained with Triplet Loss to learn a specialized embedding space. We selected the best model checkpoint at epoch 45, where training metrics showed optimal convergence: the Triplet Loss approached 0 and the training accuracy exceeded 99%, confirming the model's ability to discriminate fine-grained identities within the training distribution.

Training Results %	
mAP	81.7
Rank-1	96.5
Rank-5	98.0
Rank-10	99.0
Rank-20	99.0

4.3 Optimization Strategy

The tracking configuration underwent systematic optimization to balance identity persistence with trajectory cleanliness. To achieve this, we defined a comprehensive set of hyperparameters tailored to the dynamics of soccer.

Core Tracker Configuration: We configured the BoTSORT tracker with the following key parameters to ensure robust performance:

- **Extended Memory:** `track_buffer = 250` (10s). Allows the system to "remember" players during long absences (e.g., corner kicks).
- **Strict Association:** `match_thresh = 0.95` and `appearance_thresh = 0.90`. Ensures that re-association only happens when visual confidence is extremely high, mitigating ID swaps.
- **Spatial Proximity:** `proximity_thresh = 0.70`. Used as a fallback when visual matching is uncertain.
- **Hysteresis Thresholds:** `track_high_thresh = 0.40`, `track_low_thresh = 0.10`, `new_track_thresh = 0.15`. These values prioritize recall, allowing the tracker to maintain tracks even when detection confidence drops temporarily.
- **Advanced Features:** `fuse_scores = True` (combines detection score with association cost), `gmc_method = 'sparseOptFlow'` (for camera stabilization), `with_reid = True` (activates the fine-tuned OSNet model).

Recall-Oriented Detection Filtering: To feed the tracker with maximum information, we lowered the global detection confidence threshold to 0.20. This strategy relies on the tracker's internal logic and subsequent offline cleanup to filter noise, rather than discarding potentially valid detections prematurely.

4.4 Offline Post-Processing Pipeline

To strictly maximize trajectory continuity for the HOTA metric, we developed a custom `SoccerTrackCleaner` module. This offline refinement stage processes the raw tracking output to correct fragmentation and remove noise.

- **Track Stitching:** We implemented a stitching module configured with `max_stitch_gap=150` frames (6s) and a spatial limit of `max_stitch_dist=200` pixels. This step merges broken tracklets that exhibit spatial and temporal consistency, recovering identities lost during extended occlusions or frame re-entry.
- **Linear Interpolation:** We applied linear interpolation to fill missing detections in short temporal gaps of up to 15 frames (`max_interp_gap`). This technique effectively bridges discontinuities caused by motion blur or temporary obstructions, assuming locally linear motion.
- **Trajectory Smoothing:** We used a Savitzky-Golay filter with a window size of 15 (`smooth_window`) and a polynomial order of 2 (`smooth_poly`) to smooth the bounding box coordinates. This reduces the localization jitter typical of frame-by-frame detection, improving trajectory precision.

- **Noise Cleanup:** We removed tracks shorter than 50 frames (`min_track_len`). This aggressive filtering eliminates spurious false positives, such as flashes on billboards or artifacts, ensuring that only high-confidence trajectories contribute to the final evaluation.

4.5 Qualitative Validation

Before computing global metrics, we conducted a qualitative assessment using a side-by-side comparison video (Fig 6).

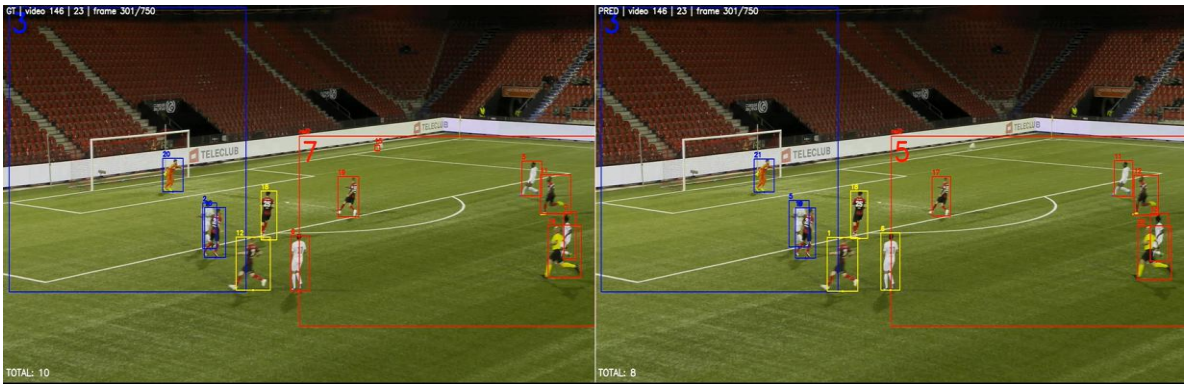


Fig. 6 : Side-by-Side Qualitative Analysis (Video 146). Left: Ground Truth (Raw Data). Right: Ours (Prediction).

The visual comparison demonstrates the semantic robustness of the proposed pipeline:

- **Enhanced Noise Filtering:** The Ground Truth (Left) includes raw annotations for the ball and out-of-bounds players. Our prediction (Right) effectively cleans the scene, filtering out non-human objects and focusing only on valid tracking targets.
- **Counting Accuracy:** While the Ground Truth displays a raw count of 7 in the Red ROI, our system computes a count of 5. This is the semantically correct value, as our model correctly excludes the ball and the boundary player (based on the P_{feet} logic), adhering strictly to the challenge's counting criteria.
- **Tracking Stability:** Unique IDs remain persistent even in dense clusters, confirming the efficacy of the BoTSORT association logic.

5. Behavior Analysis

The Behavior Analysis task requires estimating the crowd density within specific, dynamically configured Regions of Interest (ROI) for each video sequence. The primary challenge in this domain is not merely the counting process, but the accurate spatial localization of players relative to the ROI boundaries. This is significantly complicated by

the slanted perspective of television broadcast cameras, which causes 2D bounding boxes to be misleading representations of a player's physical ground position.

5.1 Methodological Approach

To accurately map 2D bounding boxes (image coordinates) onto the field geometry, we adopted a Projective Ground Point Estimation approach.

Using the standard geometric centroid (x_c, y_c) of a bounding box proved inadequate. Due to camera perspective, a player's torso might be projected inside an ROI while their feet are physically outside, or vice versa. To resolve this ambiguity and ensure physical consistency, our system considers a player "inside" an ROI if and only if their ground contact point lies within the polygon.

The formula for the *Ground Point* (P_{feet}) is defined as the midpoint of the bottom edge of the bounding box:

$$P_{feet} = \left(x_{tl} + \frac{w}{2}, y_{tl} + h\right)$$

Where (x_{tl}, y_{tl}) represent the top-left coordinates, and (w, h) are the width and height of the box. This specific point approximates the player's feet on the pitch plane, providing a more robust criterion for inclusion than IoU-based or centroid-based methods.

5.2 Performance Analysis

Quantitative evaluation was conducted using the nMAE (Normalized Mean Absolute Error) metric, which penalizes deviations between the predicted count and the Ground Truth.

Based on the experimental validation on the test split, the system achieved a Normalized Mean Absolute Error (nMAE) of 0.9455. This result indicates a high level of precision in the geometric approach. It suggests that the combination of high-resolution detection and ground-point projection effectively mitigates perspective-induced errors, providing consistent alignment with the ground truth in most scenarios.

6. Discussion and Conclusions

To establish the most effective architecture for the tracking pipeline, we conducted a comparative analysis of various object detection models, specifically evaluating YOLOv8m, YOLOv8l, YOLOv11m, and YOLOv11l. Among these, YOLOv11l demonstrated superior performance in terms of recall and precision; consequently, we selected this model and further enhanced its capabilities through specific fine-tuning on the target

dataset. Following the detection stage, we benchmarked state-of-the-art tracking algorithms, comparing BoTSORT against ByteTrack, and identified BoTSORT as the more robust solution for this application. Finally, to minimize identity switches and handle occlusions effectively, we evaluated different Re-Identification (ReID) feature extractors, including YOLOv8n-based features and OSNet_x1.0. The OSNet_x1.0 architecture yielded the most consistent results, prompting us to fine-tune this specific model to maximize discrimination accuracy and ensure tracking stability.

The experimental campaign, conducted on the provided Test Set, demonstrated the effectiveness of the proposed *Tracking-by-Detection* pipeline. The final quantitative results are exceptionally robust:

- **HOTA@0.5:** 0.8121
- **nMAE:** 0.9455
- **Final PTBS:** 1.7576

These results provide strong empirical support for our architectural choices. The exceptionally high nMAE suggests that coupling a high-resolution detector (YOLO11l @ 1920px) with a geometric projection logic (P_{feet}) effectively addresses the perspective challenges inherent in broadcast footage, offering a viable alternative to complex camera calibration. Furthermore, the HOTA score indicates the success of the hybrid Online-Offline tracking strategy: the integration of BoTSORT with robust post-processing (Stitching and Deep Interpolation) enabled the system to maintain trajectory continuity across significant temporal gaps and non-linear occlusions.